

RA8 – Dynamic Web Applications

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Theoretical Documentation

Server-side vs Client-side Execution

- **Server-side:** Code runs on the server (Laravel generates HTML pages, handles database operations, and sends fully rendered views to the client).
- **Client-side:** Code runs in the browser (JavaScript manipulates DOM, fetches data asynchronously via AJAX without reloading the page).

Role of Laravel as a Server-side Framework

- Laravel processes requests, interacts with the database, applies business logic, and generates dynamic HTML using **Blade templates**.
- It provides routing, controllers, validation, and security features out-of-the-box.

Advantages of combining Laravel with JavaScript

- Dynamic user interactions without full page reloads (AJAX).
- Improved user experience (instant feedback, form validation, delete operations).
- Separation of concerns: Laravel handles backend, JS handles frontend interactions.

Technologies Used

- **Laravel 12.51.0** – PHP server-side framework
- **Blade Templates** – Templating engine for dynamic HTML
- **HTML / CSS / Bootstrap 5** – Page structure and styling
- **JavaScript / Axios** – AJAX operations

- **MySQL** – Database storage
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Web Interface Source Code

Routes (**routes/web.php**)

```
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\ProductController;

Route::resource('products', ProductController::class);
```

Controller (**app/Http/Controllers/ProductController.php**)

```
<?php

namespace App\Http\Controllers;

use App\Models\Product;
use Illuminate\Http\Request;

class ProductController extends Controller
{
    public function index()
    {
        $products = Product::paginate(5);
        return view('products.index', compact('products'));
    }

    public function create()
    {
        return view('products.create');
    }

    public function store(Request $request)
    {
        $validated = $request->validate([
            'name' => 'required|string|max:255',
```

```

        'description' => 'nullable|string',
        'price' => 'required|numeric|min:0',
        'stock' => 'required|integer|min:0',
    ]);

    Product::create($validated);
    return redirect()->route('products.index')->with('success',
'Product created successfully.');
```

```

    }

    public function edit(Product $product)
    {
        return view('products.edit', compact('product'));
    }

    public function update(Request $request, Product $product)
    {
        $validated = $request->validate([
            'name' => 'required|string|max:255',
            'description' => 'nullable|string',
            'price' => 'required|numeric|min:0',
            'stock' => 'required|integer|min:0',
        ]);

        $product->update($validated);
        return redirect()->route('products.index')->with('success',
'Product updated successfully.');
```

```

    }

    public function destroy(Product $product)
    {
        $product->delete();
        return response()->json(['success' => true]);
    }
}

```

Model (app/Models/Product.php)

```
<?php
```

```
namespace App\Models;

use Illuminate\Database\Eloquent\Model;

class Product extends Model
{
    protected $fillable = ['name', 'description', 'price', 'stock'];
}
```

Blade Layout (**resources/views/layouts/app.blade.php**)

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Product Management</title>
    <link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css">
</head>
<body>
<div class="container mt-4">
    @yield('content')
</div>

<script
src="https://cdn.jsdelivr.net/npm/axios/dist/axios.min.js"></script>
@yield('scripts')
</body>
</html>
```

Views

Index (**resources/views/products/index.blade.php**)

```
@extends('layouts.app')

@section('content')
<h1>Products</h1>
```

```
<a href="{{ route('products.create') }}" class="btn btn-primary
mb-3">Create Product</a>
```

```
@if(session('success'))
    <div class="alert alert-success">{{ session('success') }}</div>
@endif
```

```
<table class="table table-bordered">
    <thead>
        <tr>

<th>ID</th><th>Name</th><th>Price</th><th>Stock</th><th>Actions</th>
        </tr>
    </thead>
    <tbody>
        @foreach($products as $product)
            <tr id="product-{{ $product->id }}">
                <td>{{ $product->id }}</td>
                <td>{{ $product->name }}</td>
                <td>{{ $product->price }}</td>
                <td>{{ $product->stock }}</td>
                <td>
                    <a href="{{ route('products.edit', $product) }}"
class="btn btn-sm btn-warning">Edit</a>
                    <button class="btn btn-sm btn-danger"
onclick="deleteProduct('{{ $product->id }}')">Delete</button>
                </td>
            </tr>
        @endforeach
    </tbody>
</table>
```

```
{{ $products->links() }}
@endsection
```

```
@section('scripts')
<script>
function deleteProduct(id) {
    id = parseInt(id);
    if(!confirm('Are you sure?')) return;
```

```

        axios.delete('/products/' + id)
            .then(res => {
                if(res.data.success) {
                    document.getElementById('product-' + id).remove();
                }
            })
            .catch(err => console.log(err));
    }
</script>
@endsection

```

Create (resources/views/products/create.blade.php)

```

@extends('layouts.app')

@section('content')
<h1>Create Product</h1>

@if($errors->any())
    <div class="alert alert-danger">
        <ul>@foreach($errors->all() as $error)<li>{{ $error
    }}</li>@endforeach</ul>
    </div>
@endif

<form action="{{ route('products.store') }}" method="POST">
    @csrf
    <div class="mb-3">
        <label>Name</label>
        <input type="text" name="name" class="form-control"
value="{{ old('name') }}">
    </div>
    <div class="mb-3">
        <label>Description</label>
        <textarea name="description" class="form-control">{{
old('description') }}</textarea>
    </div>
    <div class="mb-3">
        <label>Price</label>
        <input type="number" step="0.01" name="price"
class="form-control" value="{{ old('price') }}">

```

```

        </div>
        <div class="mb-3">
            <label>Stock</label>
            <input type="number" name="stock" class="form-control"
value="{{ old('stock') }}">
        </div>
        <button class="btn btn-success">Create</button>
        <a href="{{ route('products.index') }}" class="btn
btn-secondary">Back</a>
    </form>
@endsection

```

Edit (resources/views/products/edit.blade.php)

```

@extends('layouts.app')

@section('content')
<h1>Edit Product</h1>

@if($errors->any())
    <div class="alert alert-danger">
        <ul>@foreach($errors->all() as $error)<li>{{ $error
}}</li>@endforeach</ul>
    </div>
@endif

<form action="{{ route('products.update', $product) }}"
method="POST">
    @csrf
    @method('PUT')
    <div class="mb-3">
        <label>Name</label>
        <input type="text" name="name" class="form-control"
value="{{ old('name', $product->name) }}">
    </div>
    <div class="mb-3">
        <label>Description</label>
        <textarea name="description" class="form-control">{{
old('description', $product->description) }}</textarea>
    </div>
    <div class="mb-3">

```

```
        <label>Price</label>
        <input type="number" step="0.01" name="price"
class="form-control" value="{{ old('price', $product->price) }}">
    </div>
    <div class="mb-3">
        <label>Stock</label>
        <input type="number" name="stock" class="form-control"
value="{{ old('stock', $product->stock) }}">
    </div>
    <button class="btn btn-success">Update</button>
    <a href="{{ route('products.index') }}" class="btn
btn-secondary">Back</a>
</form>
@endsection
```

User Interaction Evidence

- CRUD forms for creating, editing, deleting products.
 - Flash messages (success/error) after create/update.
 - Navigation links: “Back” button, index page, create/edit forms.
-

Form Validation Evidence

- Server-side validation:

```
$validated = $request->validate([
    'name' => 'required|string|max:255',
    'description' => 'nullable|string',
    'price' => 'required|numeric|min:0',
    'stock' => 'required|integer|min:0',
]);
```


- Errors displayed in Blade views using:

```
@if($errors->any())
    <div class="alert alert-danger">
        <ul>@foreach($errors->all() as $error)<li>{{ $error
    }}</li>@endforeach</ul>
    </div>
@endif
```

Dynamic Content and AJAX Evidence

- Dynamic rendering of products from DB using `$products`.
- Pagination with `$products->links()`.
- Delete operation via **Axios** without full page reload:

```
axios.delete('/products/' + id)
    .then(res => {
        if(res.data.success) {
            document.getElementById('product-' + id).remove();
        }
    })
```

Complete Web Application Evidence

- CRUD fully functional (create, edit, delete, view list).
- Flash messages for success/error.
- AJAX operations update the list dynamically.
- Authentication & roles **not implemented**, optional for this practice.

(TE ENVIO LAS CAPTURAS Y EL VIDEO POR SEPARADO DEL DOCUMENTO PORQUE NO ME DEJA PEGAR LAS CAPTURAS EN EL DOCUMENTO NO SE PORQUE Y YA DE PASO TE AGREGO EL VIDEO)